LEAKLISTEN · DESIGN DECISIONS REGISTER

LeakListen design decisions register

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DOCUMENT CONTROL

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REVISION HISTORY

Version	Date	Author	Change
0.3	2026-11-01	Amish Chadha	Kit 1.9.1 upgrade: concept choice proposed (D10, awaiting Amish), STEM fix record (D11), modelling notes (M1 to M4); no earlier decision changed
0.2	2026-10-02	Amish Chadha	Amish approved the recommendations for all six open decisions on 2026-10-02 (LKL-DDR-003 accepted); moved to decisions made
0.1	2026-10-01	Amish Chadha	Register opened with the open decisions, items to confirm, value engineering and decisions made to date

LeakListen design decisions register

Every design decision still to be made, and every decision made, in one place. Each decision is argued in full in its decision record under `docs/decisions/`; this register is the index Amish works from. The build plan (`docs/05-build-plan.md`) describes the design as it stands and does not list open decisions.

Open decisions

Table 1. Decisions proposed and awaiting Amish.

#	Decision	Options	Recommendation	Record
D10	Concept choice	A magnet-on contact sensor (existing design, score 155 of 205); B hydrant cap hydrophone (127); C ground spike geophone (123)	Option A as the product, option B kept as the plastic-main variant; A leads B by 18 %, so no tie	LKL-DDR-004; LKL-OPT-001
D12	Optional: raise the seismic mass from 53 g to about 56 g (20 mm brass rod, 21 mm long) to bring self-noise from 1.03 to under 1.0 µg/√Hz (PR-004)	Leave as is (PR-004 stays at risk); change the dimension	Leave as is until the ceramic and amplifier data are real; the 3 g costs 3 g of the 38 g mass margin on PR-012	LKL-STM-001 S4; LKL-CAL-001 F3

To confirm when parts are bought

#	What to confirm	Why it matters	Source
1	The pot magnet's stud is M6 and can be cut to 6 mm	The puck base is tapped M6 for 6 mm of thread	LKL-DDR-003, P7
2	The flat antenna's centre stud size, and that its lead ends in an SMA plug	Sets the bracket's stud hole and the top plug's connector	LKL-DDR-003, P2 and P4
3	The SMA bulkhead's panel hole and the M12 socket's thread (M16 x 1.5 assumed)	Sets the holes in the two end plugs	LKL-DDR-003, P4 and P6
4	The square tube inserts suit 1.5 mm wall tube and take an M10 levelling foot	The feet wedge the bar	LKL-DDR-003, P1
5	The main board fits 38 x 110 mm with its components within 8 mm	The chassis and the tube bore are sized to it	LKL-DDR-003, P5
6	The capacitor and cell sizes (16 x 30 mm, C size 26.2 x 50 mm)	The chassis clips are printed to them; the cell clears the bore by 0.8 mm	LKL-DDR-003, P5

Value engineering

Value-engineering target: USD 130 (a hypothetical control target, not a limit). Estimated cost of the constructable design: USD 141 (USD 11 over the target). Main cost drivers and savings worth trying:

- The largest lines are the main board (USD 30, a prototype price; its design is TRL 4 work), the neck bar hanger (USD 18), the logger housing with its turned plugs (USD 16), the sensor cable (USD 12) and the primary cell (USD 12).
- Making the design constructable added USD 28 to the concept's USD 113: the neck bar (USD 12 more than the strap), the turned plugs, O-rings and screws (USD 4), the internal chassis (USD 4), the eye bolt and SMA bulkhead (USD 7) and the thicker puck (USD 1).
- Savings worth trying: a fixed-length bar cut to each district's neck width, with one foot and no inner tube or pin (about USD 4 and 80 g); plugs turned in batches or moulded later; a main board price that should fall once its design is done; a cable bought by the reel with a field-fit M12 plug.

Decisions made

Date	Decision	Decided by	Record
2026-09-25	D1 to D6: nightly noise-level screening; flat antenna under the cover; piezo disc sensor with the hydrophone kept for plastic networks; reuse of the FieldNode STM32WL core; C-size cell; raw samples never leave the device	Amish: "i accept all your recommendations, go with them across all repos."	LKL-DDR-001, LKL-DDR-002
2026-09-25	D7: measure cover loss first, then a gateway within about 0.5 km of each district with iron covers	Amish, same instruction	LKL-DDR-002
2026-09-25	D8: the contact sensor's scope is metallic mains; the hydrophone variant answers plastic mains	Amish, same instruction	LKL-DDR-002
2026-09-25	D9: 42 mm pot magnet with a keeper plate	Amish, same instruction	LKL-DDR-002
2026-10-01	Budget treated as a value-engineering target, reported as over or under rather than met or not met	Amish: "the budgets are a hypothetical control target to ensure we are thinking along a value engineering lens"	This register; LKL-CAL-001 v0.3
2026-10-02	Design for construction accepted as a whole: the changes P1 to P9 of LKL-DDR-003 (neck bar hanger, flanged end plugs, top plug fittings, internal chassis, puck changes) and their knock-on changes, as made	Amish: "i approve your recommendations for all 555 open decisions."	LKL-DDR-003, P1 to P9
2026-10-02	R12 restated as 1.0 kg for the logger, cable and sensor (0.96 kg estimated); the 0.32 kg neck bar is site hardware that stays in the chamber; the logger set is weighed at TRL 4 because the margin is 0.04 kg	Amish: "i approve your recommendations for all 555 open decisions."	LKL-DDR-003, A1
2026-10-02	Neck bar for the prototype; the pilot utility is asked for its range of chamber neck sizes before any deployment, and a longer inner tube is designed only if that range goes past 710 mm; a drilled bracket stays the fallback for poor or out-of-range necks	Amish: "i approve your recommendations for all 555 open decisions."	LKL-DDR-003, A2
2026-10-02	First partner: a water utility with metallic mains, district metered areas and a leakage target it reports against. First candidate region to approach: the UK, where water companies run district metered areas and report leakage to the regulator; that would set the radio band to EU868	Amish: "i approve your recommendations for all 555 open decisions."	LKL-DDR-002, O1
2026-10-02	The vibration calibration check goes in CalRig as a small shaker bay; LeakListen defines the acceptance test (frequency range, level and pass band) in its own repo	Amish: "i approve your recommendations for all 555 open decisions."	LKL-DDR-002, O2
2026-10-02	Light pipe on the top plug kept, lit only for a brief blink at power-up or magnet swipe and added to BOM line 7; board orientation, cable route and render context accepted as appearance only	Amish: "i approve your recommendations for all 555 open decisions."	`docs/REVIEW.md`, 2026-09-26
2026-11-01	D11 (decided under Amish's standing instruction of 2026-10-23): the STEM check row S3 failed in a detail, a contact sensor hears the reference leak only about 4 m along PVC. The fix keeps the working principle and was already decided by Amish on 2026-09-25 (D8: contact sensor for metallic mains only); the hydrophone variant that answers plastic mains is now	Standing instruction, 2026-10-23, and D8	LKL-STM-001

Date	Decision	Decided by	Record
	modelled at D3 as option B and screened in the STEM check (SB1). No design dimension changed.		
2026-11-01	M1 to M4: modelling notes, no dimension changed. M1: cad/src/model.py rebuilt to the kit 1.9 interface; the neck bar is now 0.31 kg (was 0.32 kg) because its tubes are real stock sections with holes, so the set is 1.34 kg and the logger, cable and sensor stay 0.96 kg. M2: the eye bolt is M6 x 30 (BOM line 13 said M6 x 20) so that it clears the plug and washer; the BOM text is updated. M3: the M12 coupling nut is modelled engaged 10 mm on the socket thread. M4: materials missing from the kit library (PVC, acetal, nylon, FR4, nitrile and others) are added in cad/src/lkl_parts.py.	Design lead, design-intent modelling	cad/src/lkl_parts.py; docs/REVIEW.md